

HULAT-UC3M at TSAR 2025 Shared Task: **A Prompt-Based Approach using Lightweight Language Models for** **Readability-Controlled Text Simplification**

Jesus M. Sanchez-Gomez, Lourdes Moreno, Paloma Martínez, Marco Antonio Sánchez-Escudero
Computer Science and Engineering Department, Universidad Carlos III de Madrid, Spain
{jesusmsa, lmoreno, pmf, marcoasa}@inf.uc3m.es

Motivation

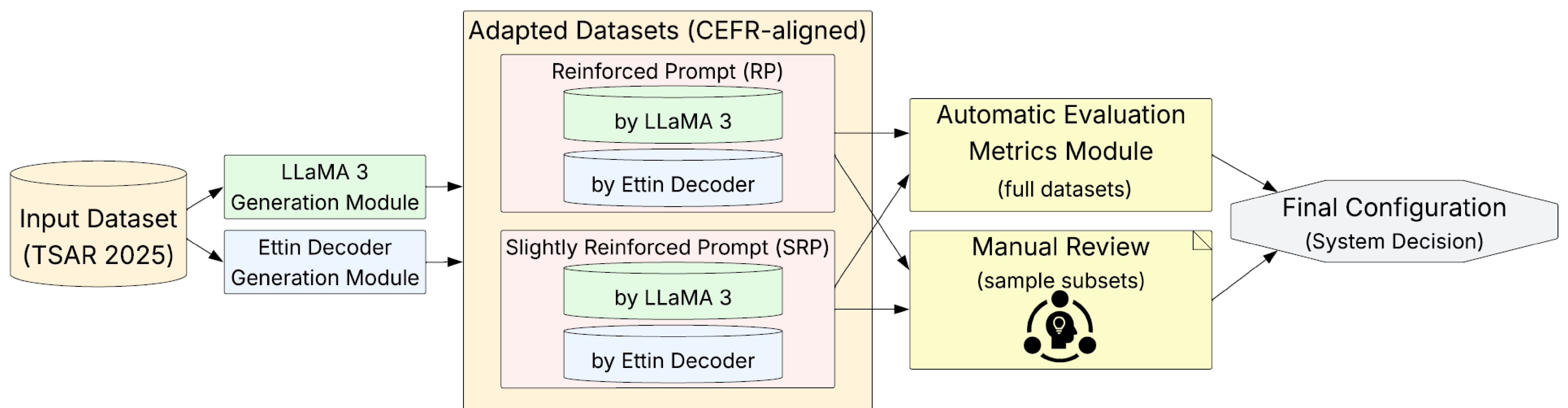
- Large Language Models (LLMs) often struggle to apply specific reading levels like CEFR consistently.
- LLMs tend to generate outputs that are too complex, frequently over-predicting intermediate levels (B1/B2).
- We need better methods to control text complexity and more reliable metrics for factual correctness in simplification.

Objectives

- To test lightweight, open-source LLMs like LLaMA 3 and Ettin Decoder.
- To compare the effectiveness of prompt-engineering strategies for CEFR-level control.
- To evaluate if a zero-shot approach can perform well without resource-intensive fine-tuning.

Proposed Architecture

- The system uses two lightweight LLMs: LLaMA 3 8B and Ettin Decoder 400M.
- Two prompt strategies were designed: a detailed reinforced prompt and a concise, slightly reinforced prompt.



Experimental Setup and Evaluation

- Trial data for system testing.
- Official metrics: CEFR Compliance, Meaning Preservation, and Similarity to References.
- Complementary metric: AlignScore, used to assess factual consistency and detect hallucinations.
- Manual review: qualitative categorization of errors such as hallucinations, minimal simplification, repetition, incoherence, oversimplification, robotic tone, and overloaded outputs.

Results and Discussion

- Slightly reinforced prompt (SRP) outperformed the reinforced one (RP), reducing complexity errors and improving factual consistency.
- LLaMA 3 8B with SRP was the most balanced model, achieving the top AlignScore.
- Ettin Decoder outputs (RP/SRP) were unreliable due to hallucinations and repetitions despite high similarity scores.
- LLaMA 3 showed the usual LLM bias of overshooting the target CEFR level (A2 often rated as B1).

Metric	LLaMA 3 8B (RP)	LLaMA 3 8B (SRP)	Ettin Decoder 400M (RP)	Ettin Decoder 400M (SRP)
CEFR Compliance - Weighted F1	0.3000	0.5200	0.4800	0.2800
CEFR Compliance - RMSE	1.1100	0.7746	0.8900	1.1619
Meaning Preservation - MeaningBERT	0.6532	0.7170	0.6901	0.8176
Meaning Preservation - BERTScore	0.8837	0.8999	0.9025	0.8807
Similarity to References - MeaningBERT	0.6384	0.7075	0.6243	0.6199
Similarity to References - BERTScore	0.8764	0.8921	0.8789	0.8470
AlignScore	0.5600	0.6038	0.4300	0.5576

Table 1: Results of experiments with trial data (RP means reinforced prompt, and SRP slightly reinforced prompt).

Model	Position	RMSE	MeaningBERT-orig	MeaningBERT-ref	AvgScore	AUTORANK
LLaMA 3 8B (RP)	31	0.682	0.790	0.791	-0.122	10.560
LLaMA 3 8B (SRP)	24	0.608	0.793	0.806	-0.028	9.370

Table 2: Results obtained in TSAR 2025 (RP means reinforced prompt, and SRP slightly reinforced prompt).

ACKNOWLEDGEMENTS. This work has been supported by grant **PID2023-148577OB-C21** (Human-Centered AI: User-Driven Adapted Language Models-HUMAN_AI) by MICIU/AEI/10.13039/501100011033 and by FEDER/UE, and by grant **2024/00752/001** (Ayudas para la Actividad Investigadora de los Jóvenes Doctores, del Programa Propio de Investigación de la UC3M).